Final Presentation

Project 3: University Library Use in Health Sciences and STEM Fields

University of Michigan in Ann Arbor

Adele Chase, Javier Martínez Rodríguez, Xiao Zeng

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Project Overview

01

U-M Library Campus-Wide Survey

- Conducted in the winter of 2023-24.
- It included faculty members of all types (e.g., instructional, clinical, research), graduate students, house officers, and staff members.
- The survey asked about the use of library spaces, physical and electronic materials of all types, and services such as consultation, document delivery, library workshops, and more.

The Dataset



Distribution - Participants in Health Sciences and STEM

Project Objective

To describe in a comprehensive manner how those in health sciences and STEM fields experience the academic library, and how the academic library can meet newly-identified needs of its user populations.



Individual Projects

02

Adele Chase

Research Question

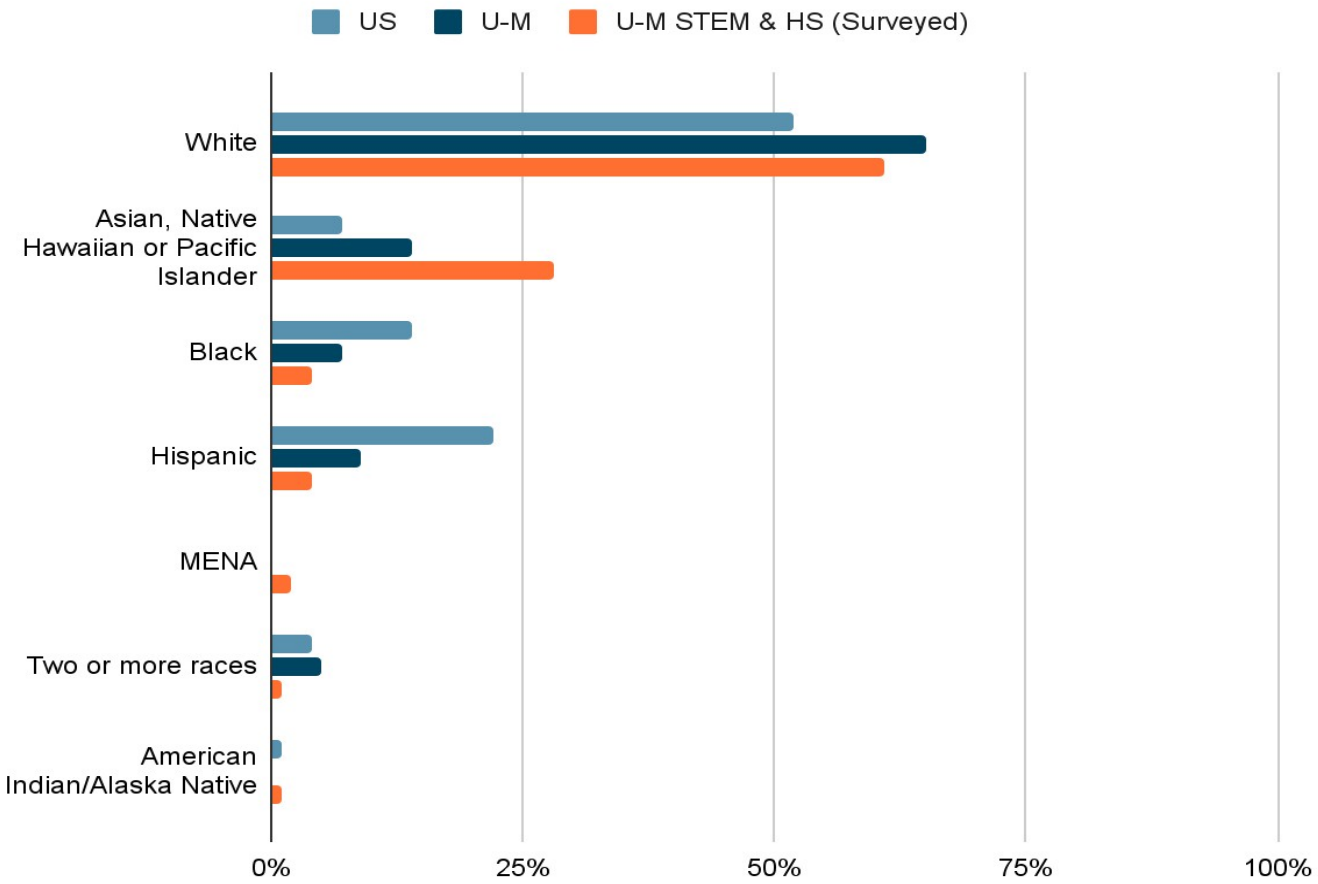
Are there differences in how faculty, researchers, and graduate students of different racial backgrounds in the health sciences and STEM perceive and use the library, its collections, research support, and instruction support?

Objective

To identify areas where UMich Libraries can implement more inclusive policies and services

Overview

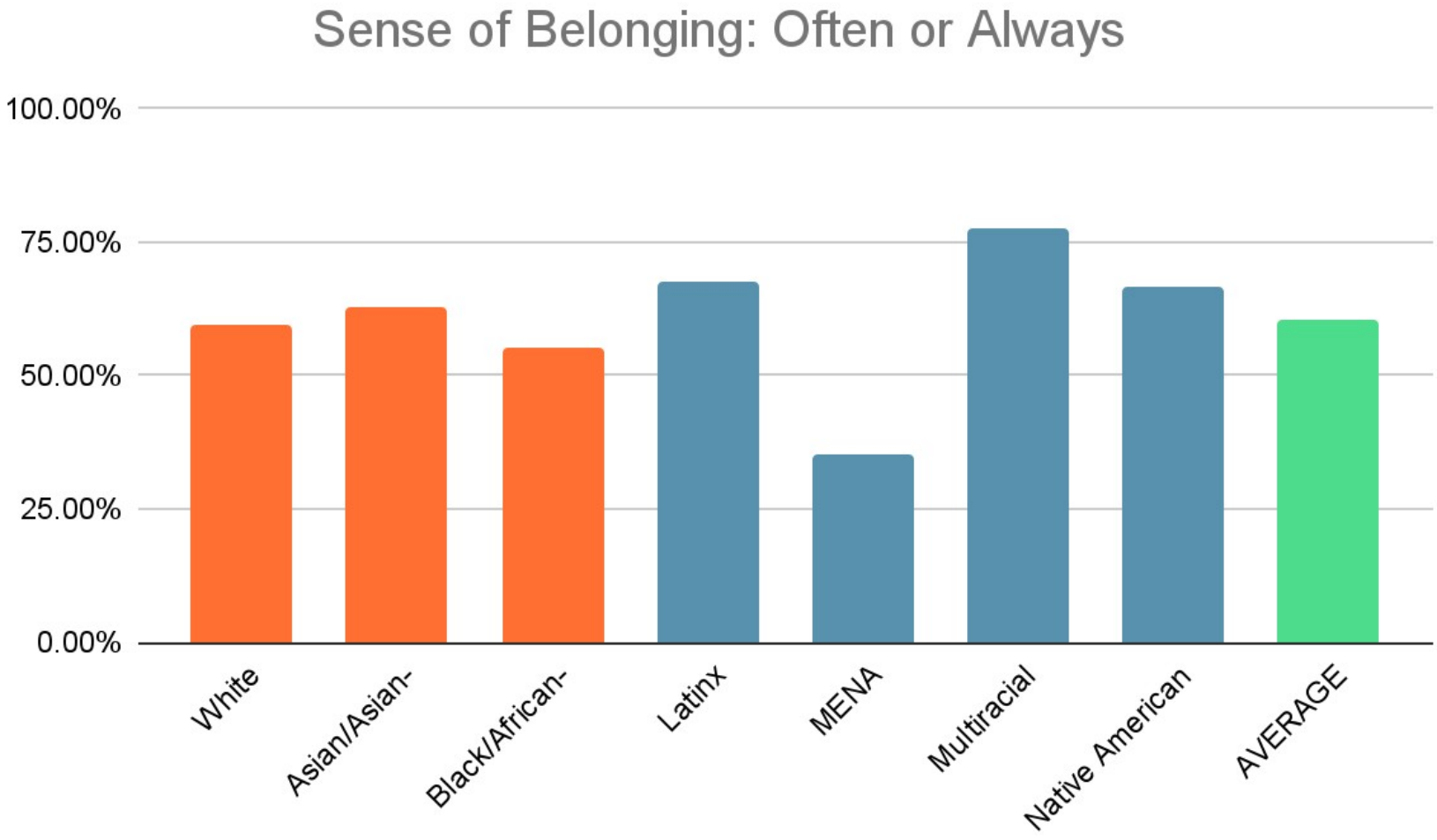
Racial-Ethnic Breakdowns

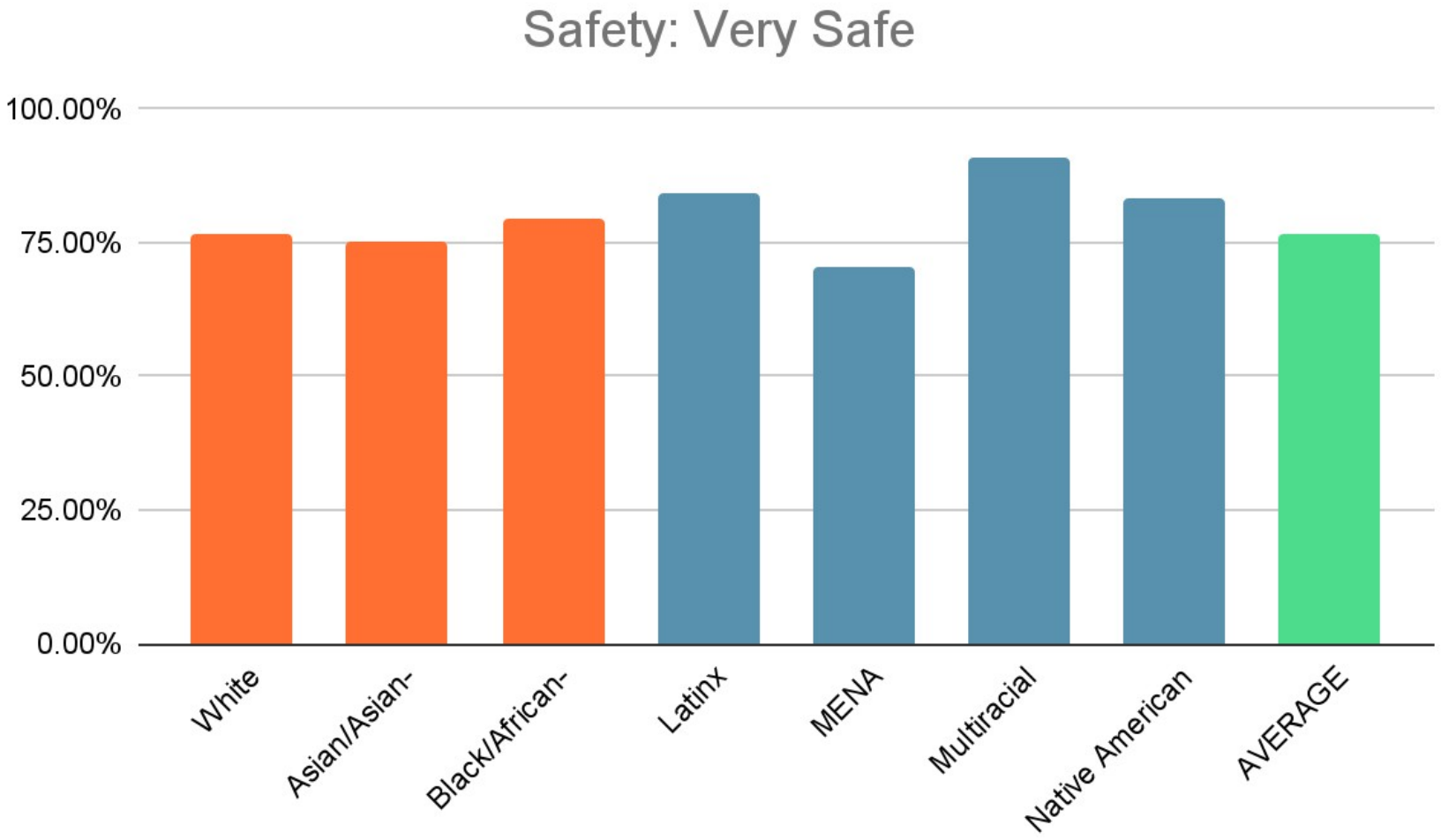


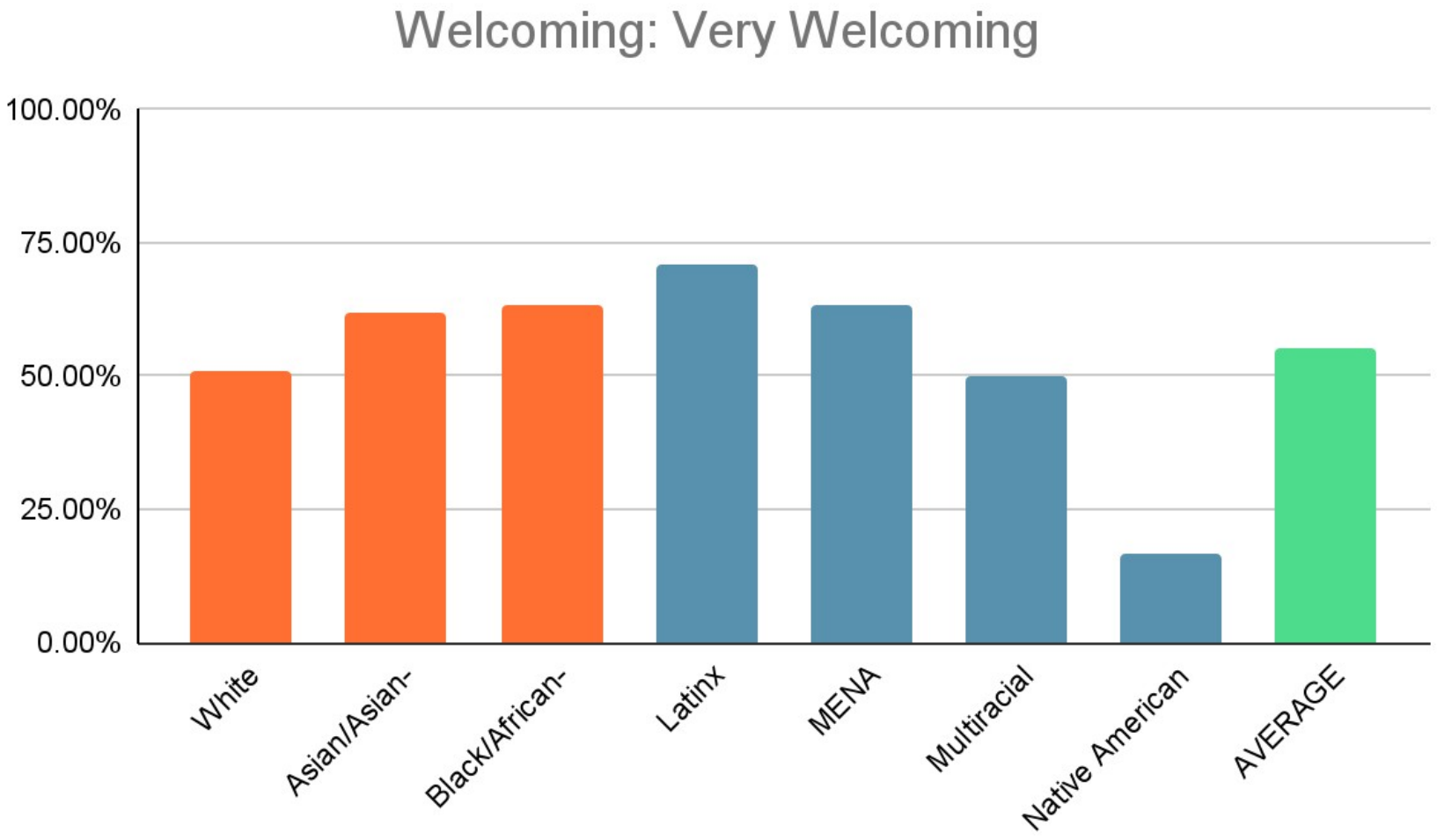
Results

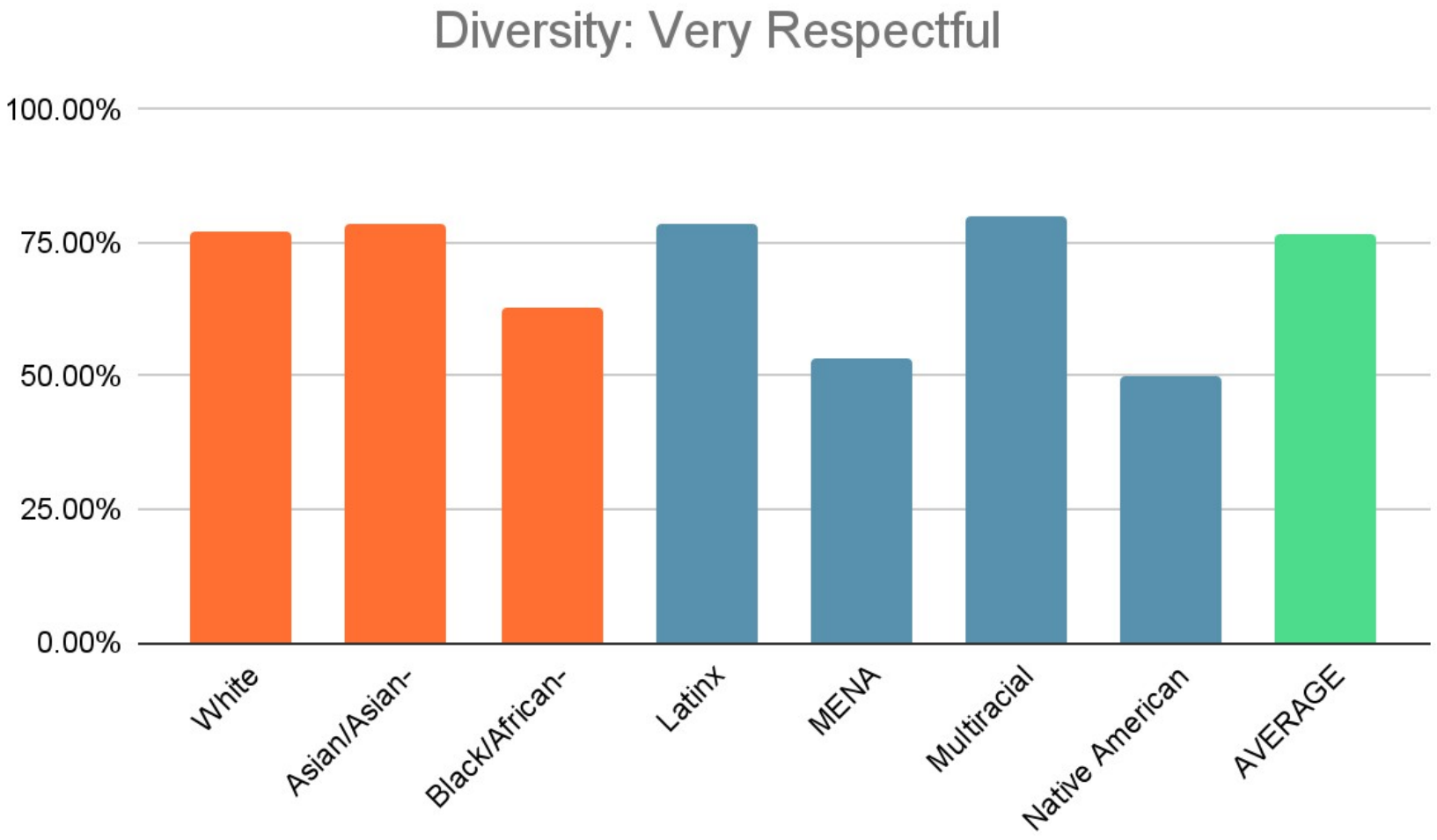
General Perceptions

Belonging, Safety, Welcoming, and Diversity


Sense of Belonging: Often or Always
100.00%
75.00%
50.00%
25.00%
0.00%
White
Asian/Asian-
Black/African-
Latinx
MENA
Multiracial
Native American
AVERAGE


Safety: Very Safe
100.00%
75.00%
50.00%
25.00%
0.00%
White
Asian/Asian-
Black/African-
Latinx
MENA
Multiracial
Native American
AVERAGE


Welcoming: Very Welcoming
100.00%
75.00%
50.00%
25.00%
0.00%
White
Asian/Asian-
Black/African-
Latinx
MENA
Multiracial
Native American
AVERAGE


Diversity: Very Respectful
100.00%
75.00%
50.00%
25.00%
0.00%
White
Asian/Asian-
Black/African-
Latinx
MENA
Multiracial
Native American
AVERAGE

Discussion: General Perceptions

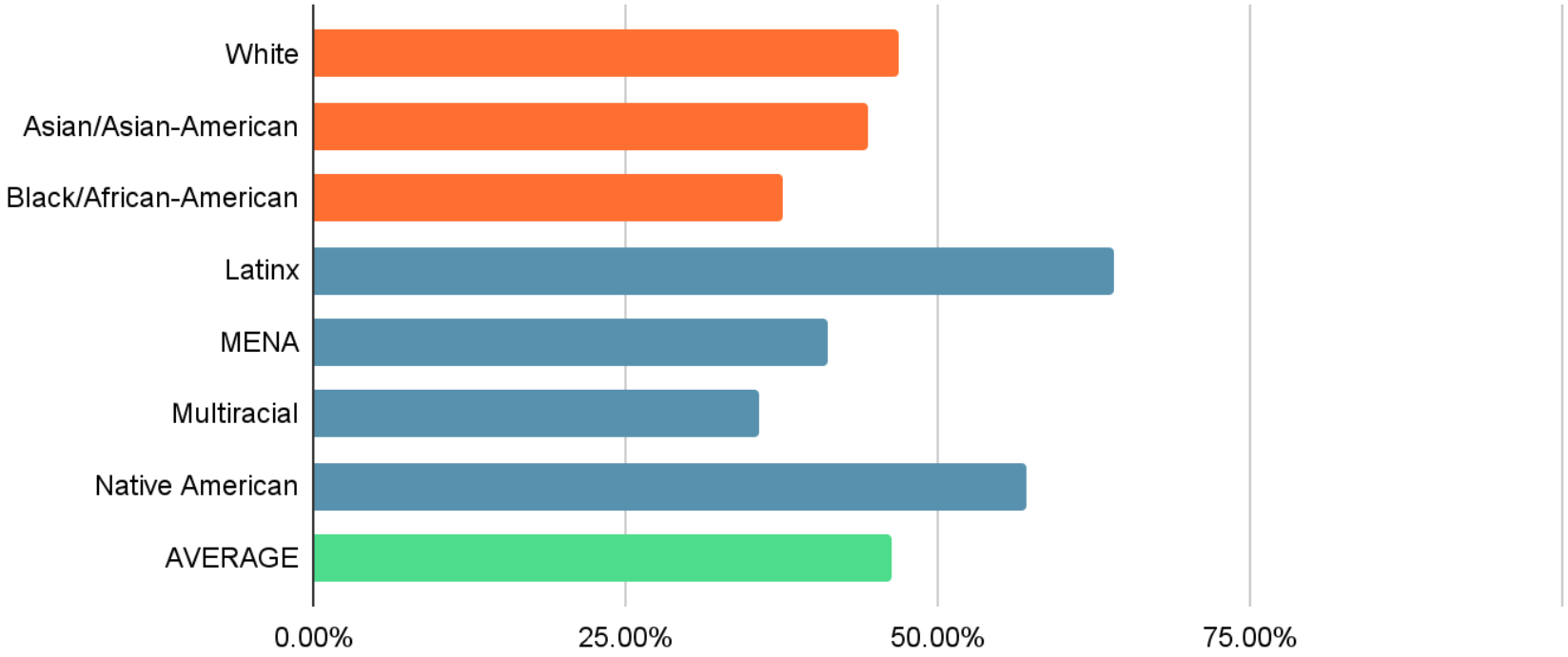
- Averages were high across the board:
  - 60.4% always or often felt a sense of belonging.
  - 76.8% felt very safe.
  - 55.3% felt that the libraries are welcoming.
  - 76.6 felt the libraries are very respectful of diversity
- A lower percentage of Black/African-American respondents felt the libraries are very respectful of diversity.
- A lower percentage of White library users felt that libraries were very welcoming.

How does gender affect perceptions?

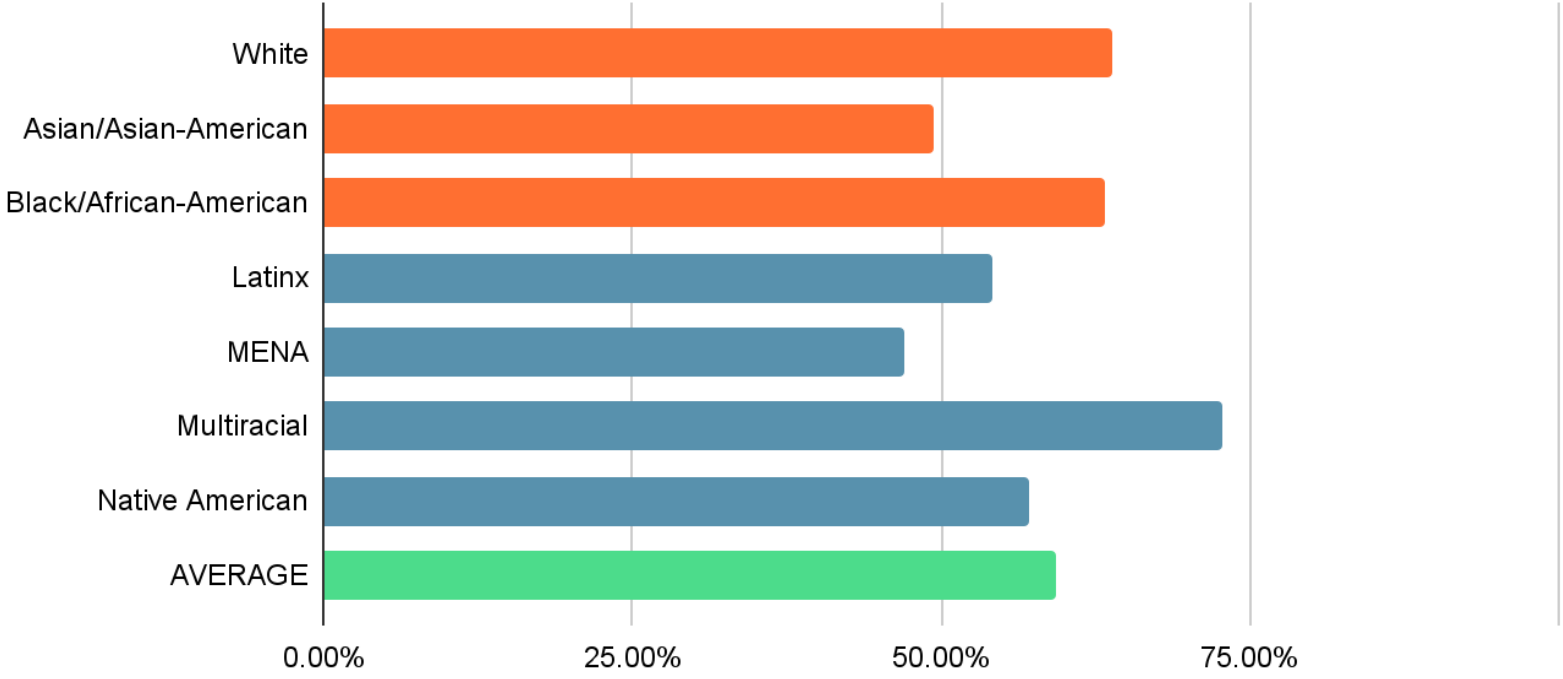
- Males have a slightly higher sense of belonging than females, with the largest gender gap among Black/African-American respondents.
- Males feel significantly safer than females, indicating a notable gender disparity in perceived safety, particularly among White respondents.
- Males feel more welcomed than females, with the largest gap seen among Black/African-American respondents.
- Females have a more positive view of the library's respect for diversity compared to males, with a significant gap between male and female Black/African-American respondents.

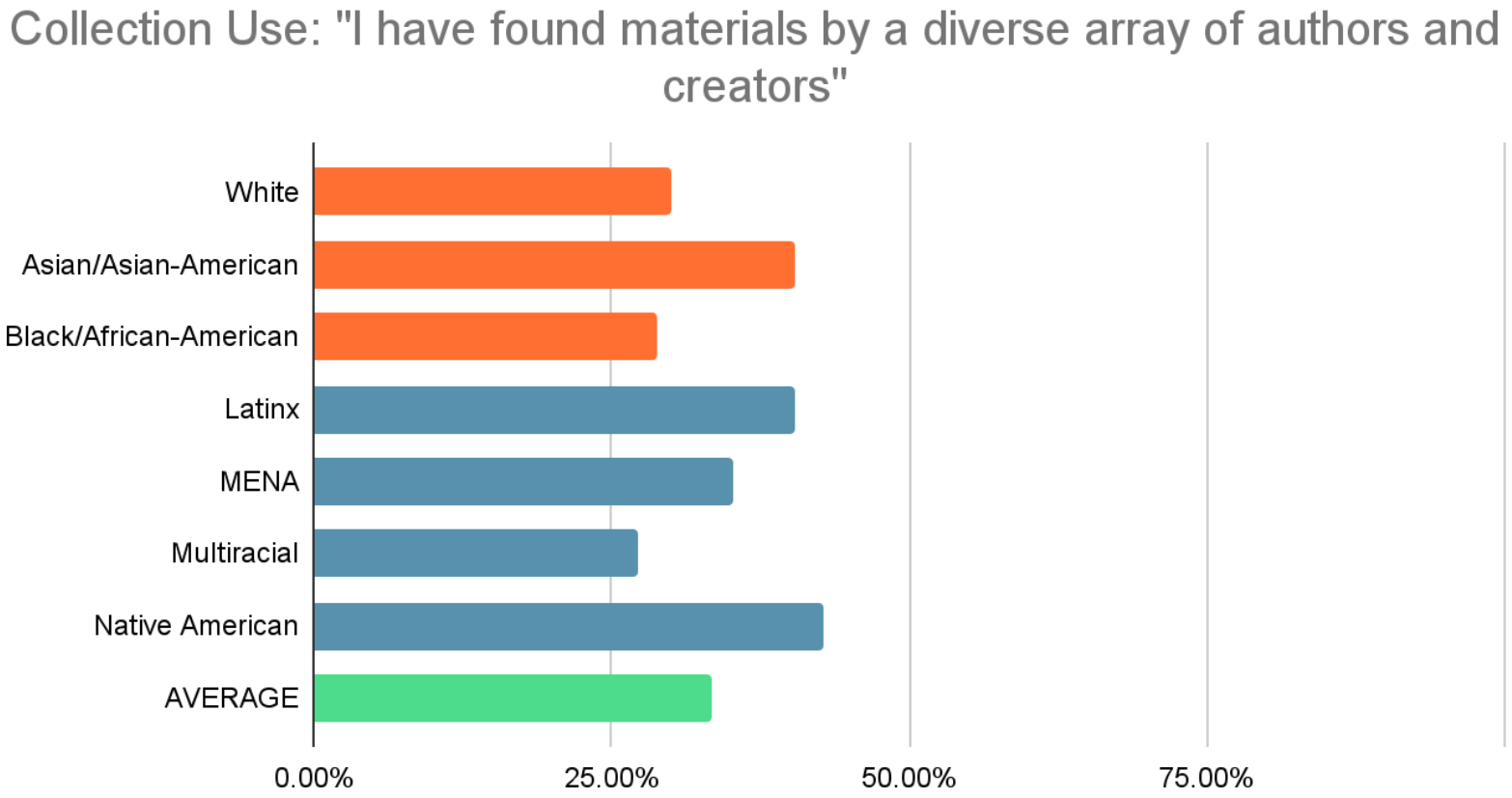
Collection use, Instruction Support, and Research Support

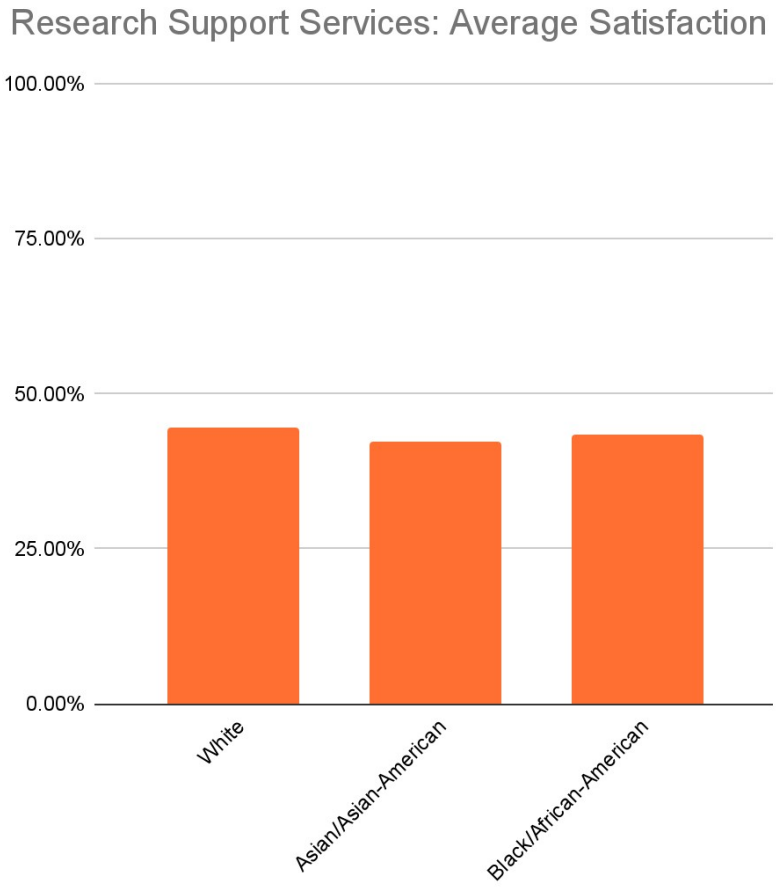
Collection Use: Materials and Resources in the Collection Always or Almost Always Meet My Needs



Collection Use: "I don't have enough information to answer the question about how the library is doing in terms of collecting diverse materials"

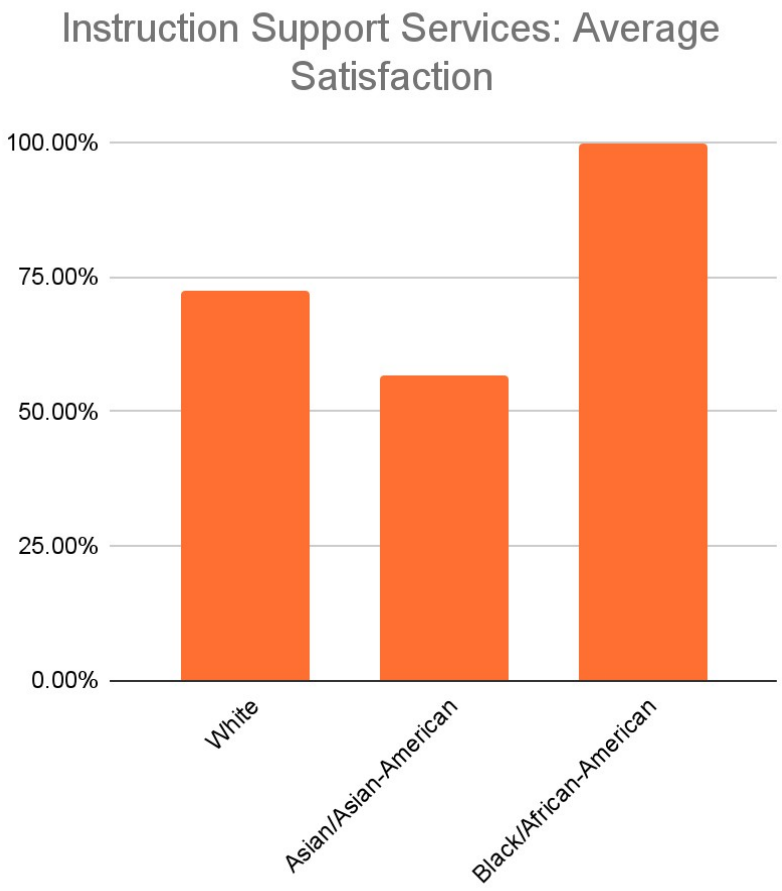



Collection Use: "I have found materials by a diverse array of authors and creators"
White
Asian/Asian-American
Black/African-American
Latinx
MENA
Multiracial
Native American
AVERAGE
0.00%
25.00%
50.00%
75.00%



This averages the percentage of each racial group who reported the highest satisfaction level for the following research support services:

- Consultation with experts or subject specialists
- Digital scholarship
- Online research guides
- Funding and grants info
- Research impact
- Deep Blue
- Publishing
- Data visualizations



This averages the percentage of each racial group who reported the highest satisfaction level for the following instruction support services:

- Help designing curriculum
- Course-relevant library services information
- Group instruction
- Tailored student consultations
- Information evaluation guidance for students
- Providing tools or materials

Discussion: Collection Use, Research Support, and Instruction Support

- The results indicate a need to improve research support services.
  - Mixed satisfaction with digital scholarship service, data visualization support.
- The results are broadly similar across racial groups, but there were very few responses for each of these questions, especially the questions on research support and instruction support.

Key Findings

“Inclusion for All” Framework

When implementing diversity initiatives, it is important to minimize backlash from dominant groups, which negatively impacts people from intentionally marginalized backgrounds.

Goal: To achieve inclusion without triggering reactance from dominant groups.

Source: Brannon, T. N., Carter, E. R., Murdock-Perriera, L. A., & Higginbotham, G. D. (2018). From backlash to inclusion for all: Instituting diversity efforts to maximize benefits across group lines. Social issues and policy review, 12(1), 57-90. https://doi.org/10.1111/sipr.12040

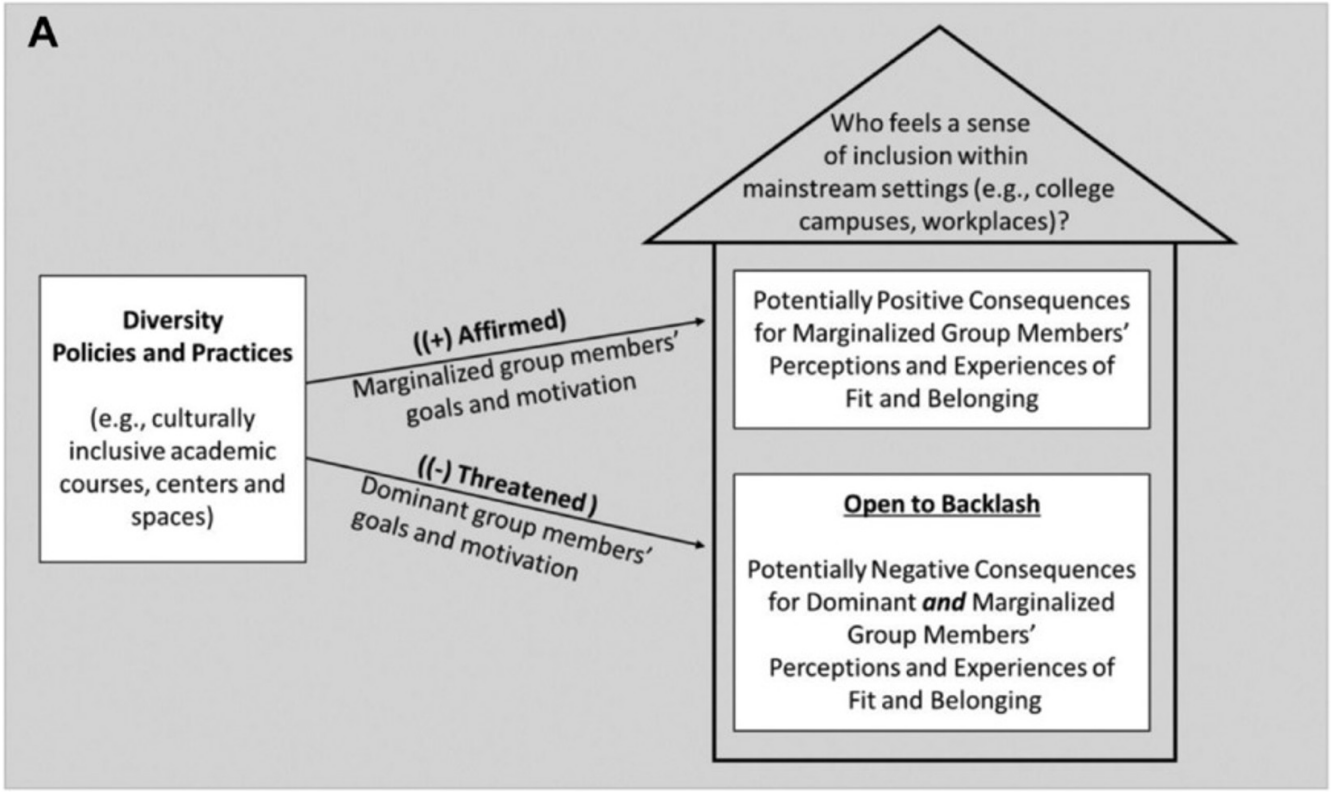
Evidence of backlash to diversity Initiatives?

- Some evidence of lower engagement or satisfaction among white respondents on two key items:
  - Whether the library has a welcoming atmosphere
  - Finding materials by creators from diverse backgrounds
- Some evidence of lower engagement or satisfaction among Black/African-American respondents on the diversity of materials in the collection
- No strong evidence of widespread backlash, but these numbers are worth keeping an eye on in the future.

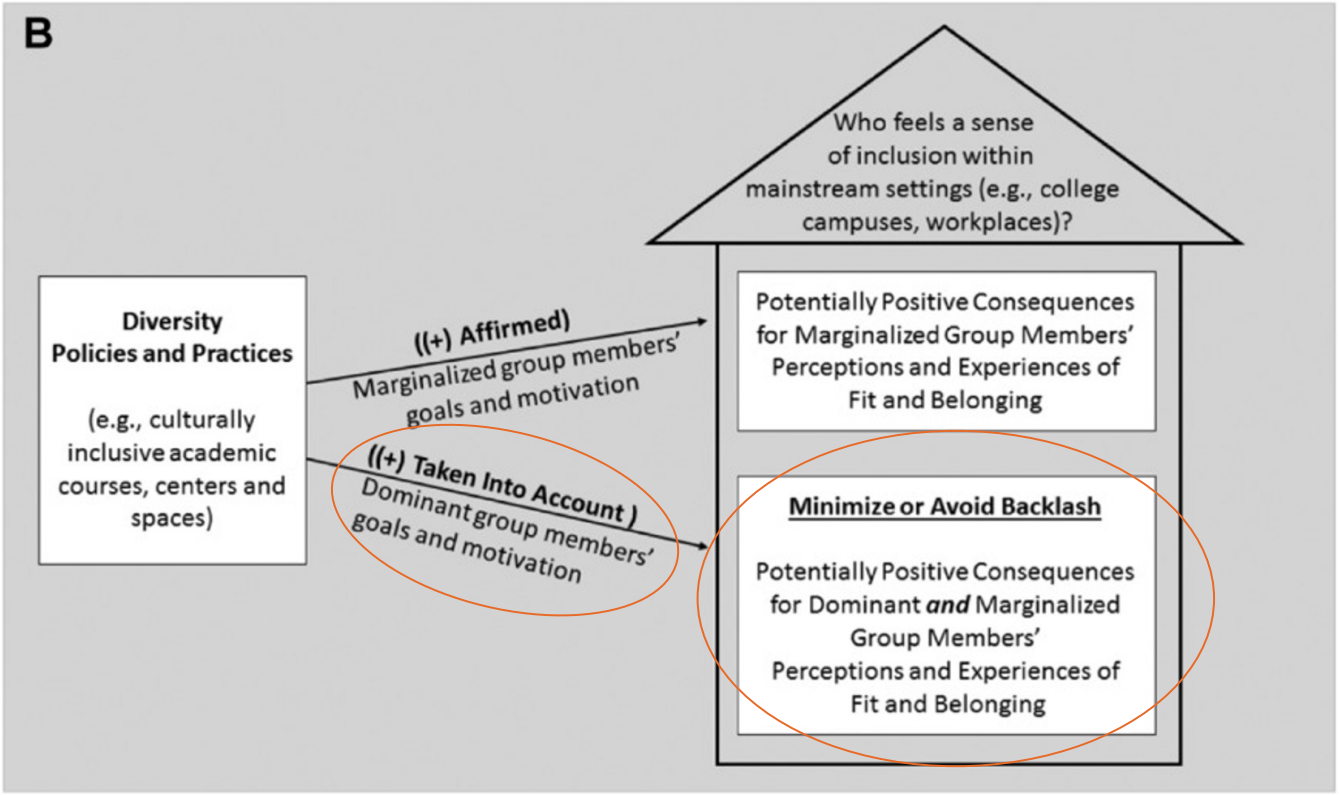
Sources of backlash and how to address them:

- Perceived Restriction of Independence
  - Solution: Provide choice and autonomy
- Preference for the Status Quo and Colorblindness
  - Solution: Structured environments for intergroup interactions
- Beliefs that Racial and Other Social Equalities Have Been Reached
  - Solution: Teach background on racial inequality. Emphasize fairness and legitimacy of diversity efforts

“Inclusion for All” Framework



“Inclusion for All” Framework



Javier Martínez Rodríguez

Question and Objective

Question

What existing library services can be leveraged to promote sponsored research in Medical and Health Sciences at the University of Michigan in Ann Arbor?

Objective

General Objective: to identify library services that can serve as a gateway for increased visibility of services to seek research funding.

The library is “a key campus player in supporting research activities throughout the research life cycle.”

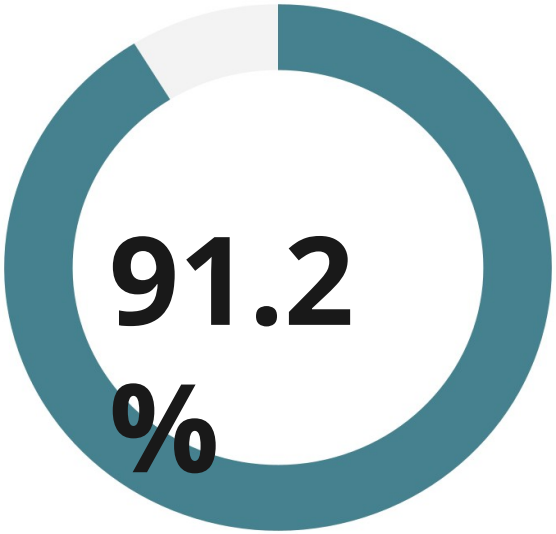
(Bryant et al., 2020, p.12)

“The campus library can serve as a central information resource [...], providing information to connect funding needs with potential funders.”

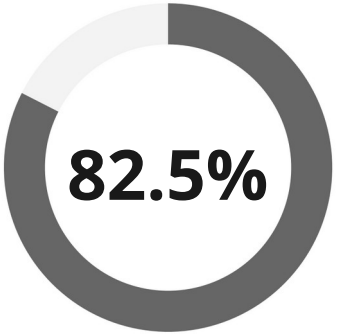
(Downing, 2009, p. 5)

Librarians and researchers agree: funding seeking consultations are a valuable service for the library to provide but there can be a lack of knowledge on the library’s capacity to do this.

(Vaughan et al., 2013)



of researchers in HS-STEM who use library resources to seek funding have published their work.



Context

- **Research Funding and Grants Guide.**
  - “campus-wide solution to help faculty locate funding resources, identify grant-writing training opportunities, and locate special events focused on finding funding opportunities.” (Rosenzweig et al., 2016, p.52)
- **Funding Search Consultations**
  - “offer personalized, expert funding and grant search consultation to University of Michigan faculty and staff.” (MLibrary, University of Michigan, 2024).

What other library services can be leveraged to increase visibility of these existing resources?

Methodology

Type of Research:

Quantitative Analysis of data from the U-M Library Campus-Wide Survey.

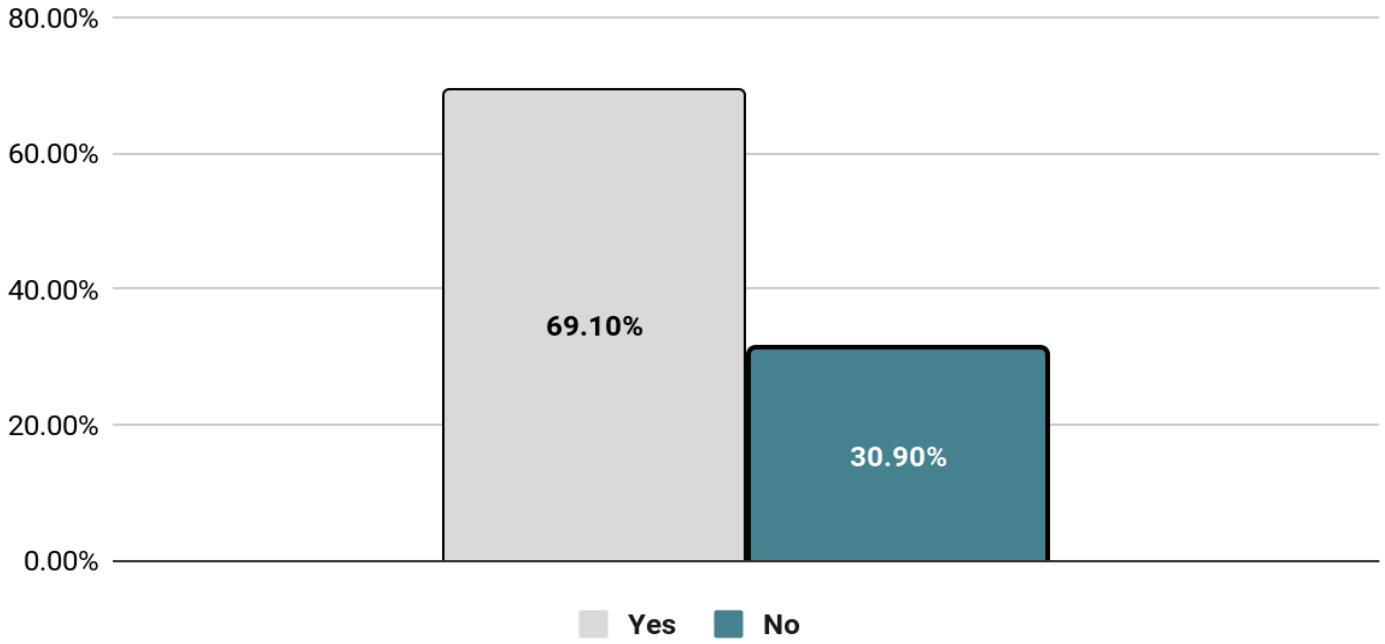
Data delimitation:

- “What is your most primary role at U-M?”
  - Selected one of the following: “Faculty member, Graduate student, Postdoctoral fellow, Visiting researcher or scholar, House officer.”
- During the past two years, have you been the leader of an original research or creative project? (As a principal investigator, a project director, a lead author, a dissertation writer, a project grant awardee, etc.)
  - Selected “Yes”.
- “What broad category (or categories) best describe your research and academic work?”
  - Selected “Medical and Health Sciences”.
- “In your research and creative work, do you engage in any of the following activities?”
  - Selected “Seeking funding”.

Results

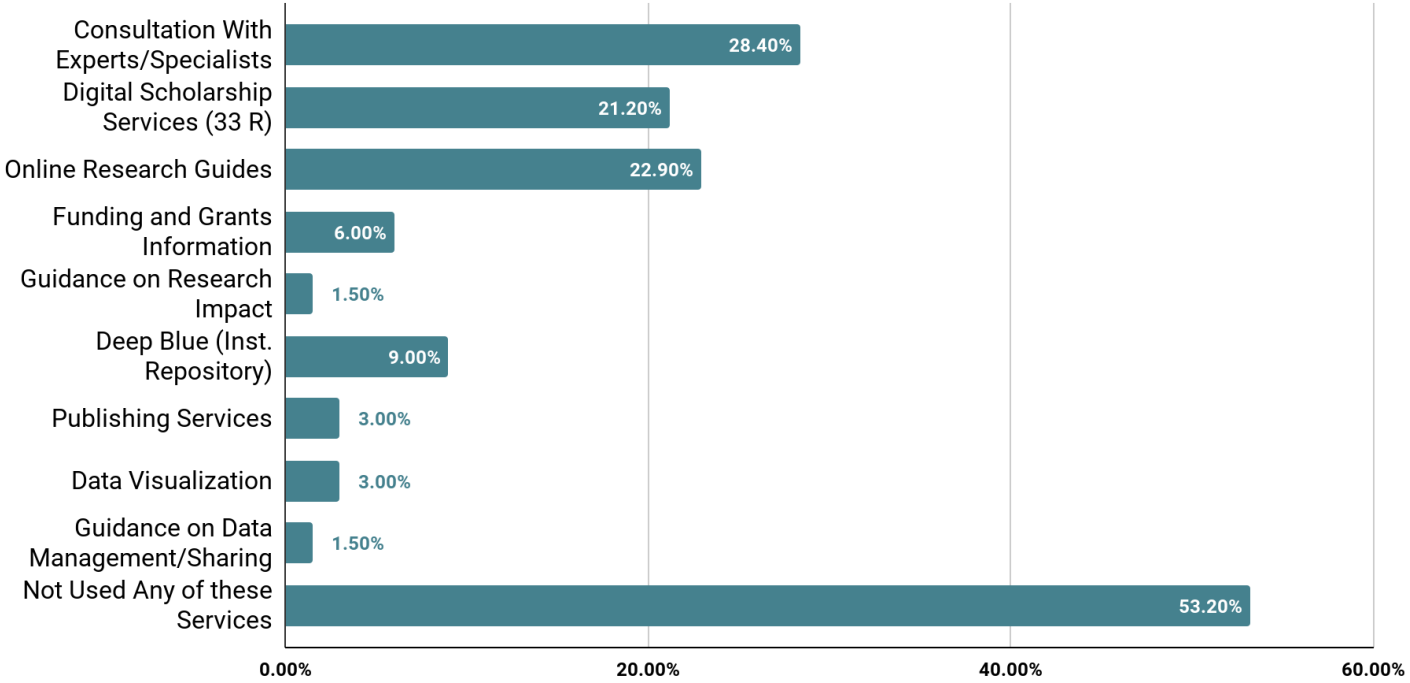
Activity: Using Library Services to Seek Funding

Frequencies (670 Respondents)



Library Research Services by Researchers Not Engaging with Services to Seek Funding

Frequencies (201 Respondents)



Insights

	Observation	Recommendation
1	Consultations with Experts/Specialists is the most frequently used service.	Exploring this service as a gateway to promoting/increasing visibility for grant seeking assistance services.
2	Those who do not use services to seek funding, do not use any research services at the library.	Outreach to these participants to increase access to all library services is necessary.

Important consideration: the ability to extrapolate these observations to the rest of Health and Medical Sciences community at U-M is limited by the number of respondents left after data delimitation.

Challenges

- Becoming familiar with SPSS.
- Considering all the different ways to approach this topic and narrowing down to a feasible project.
- Managing the timeline for this project with competing priorities.

Successes

- I was able to develop observations based on data analysis.
- I was able to create data visualizations to illustrate my arguments.
- I have created connections with members of my project team and the team at U-M.
- I have a new understanding of my strengths and limitations as a future data professional.

Next Steps

- Continue to learn about statistics.
- Continue to learn about data tools.
- Reflect, in-depth, on my experience and consider how this can be submitted for conferences or other resource sharing opportunities.
- Continue to grow my network.

References

Bryant, R., Dortmund, A., & Lavoie, B. F. (2020). Social Interoperability in Research Support: Cross-campus Partnerships and the University Research Enterprise. OCLC Research.

Downing, K. (2009). Show Me the Money! An Academic Library's Role in the Art of GrantSeeking on Campus and in the Community. https://alair.ala.org/server/api/core/bitstreams/0e0d2727-2ab1-4f73-91f5-f083ba1938ea/content

MLibrary, University of Michigan. (2024, February 8). Find Funding - Research Funding and Grants Guide - Research Guides at University of Michigan Library. Research Guides. Retrieved July 29, 2024, from https://guides.lib.umich.edu/c.php?g=283405&p=1887268

Rosenzweig, M., Smith, J. E., Curtis, A., & Puffenberger, A. (2016, Jan 21). Simplifying the Process for Finding Research Funding: A Cross-Campus Collaboration at a Large Academic Institution. Medical Reference Services Quarterly, 35(1), 52-63. 10.1080/02763869.2016.1117289

Vaughan, K. T., Hayes, B. E., Lerner, R. C., McElfresh, K. R., Romito, D., Reeves, L. H., & Morris, E. N. (2013, Oct). Development of the research lifecycle model for library services. Journal of the Medical Library Association : JMLA, 101(4), 310-314. 10.3163/1536-5050.101.4.013

Xiao Zeng

Research Question

How can the U-Mich Library address information-seeking behavior issues and make improvements to better support the interdisciplinary research needs in the Health and STEM fields, bridging the gap between researchers engaged in higher interdisciplinary studies and those involved in fewer interdisciplinary studies?

RQ 1. What are the common information-seeking behaviors among researchers in interdisciplinary studies in Health and STEM fields?

RQ 2. What issues do these researchers encounter in accessing and utilizing library resources?

RQ 3. What improvements can be made to library services to better support interdisciplinary research in these fields?

Objective

The objective of this research is to propose improvements that can enhance the library's support for interdisciplinary research and bridging the gap between different numbers of interdisciplinary studies.

Methodology

Research Method: Mixed Method

Tools:

1. SPSS (quantitative data & qualitative free text response)
2. Google Sheet
3. Google Looker Studio

Dataset:

4. Delimited Dataset
5. Created crosstabs
6. Conducted Chi-Square and Correlation tests
7. Analyzed qualitative survey responses.

Frequency of Interdisciplinary Research Respondents

Frequency_Interdisciplinary ▲	Frequency	Percent
2	335	75.3
3	91	20.4
4	15	3.4
5	3	0.7
6	1	0.2

445/15,116 Survey Results

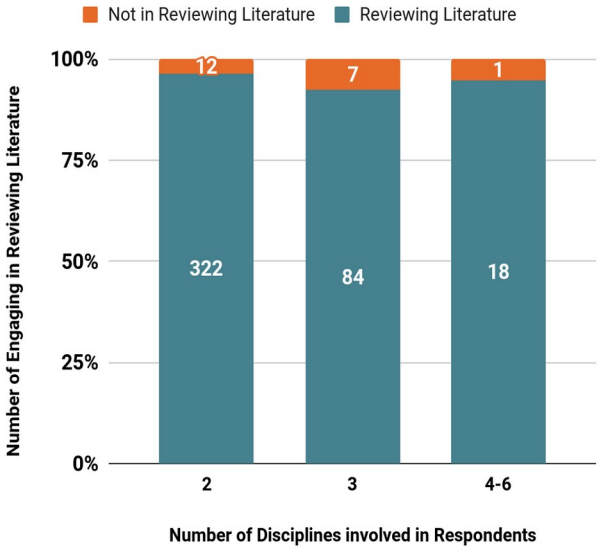
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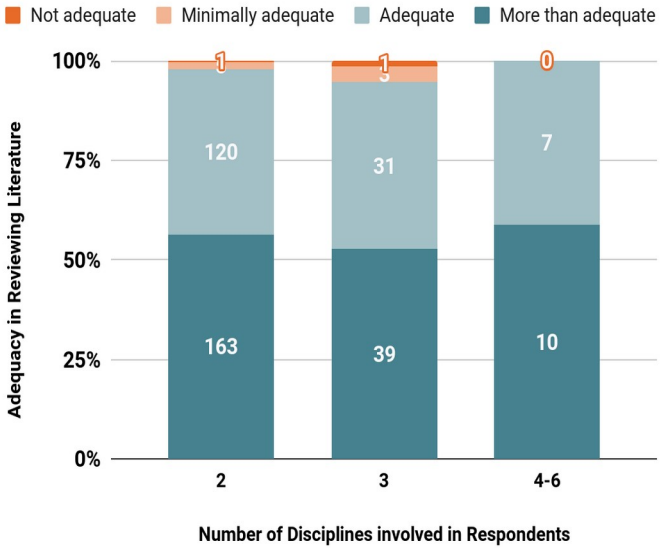
- In Engineering, Natural Sciences, and Medical and Health field.
- Involved in research in recent two years

Inverse relationship between number of disciplines involved and the number of respondents.

RQ 1. What are the common information-seeking behaviors among researchers in interdisciplinary studies in Health and STEM fields?

Research Activities: Reviewing Literature



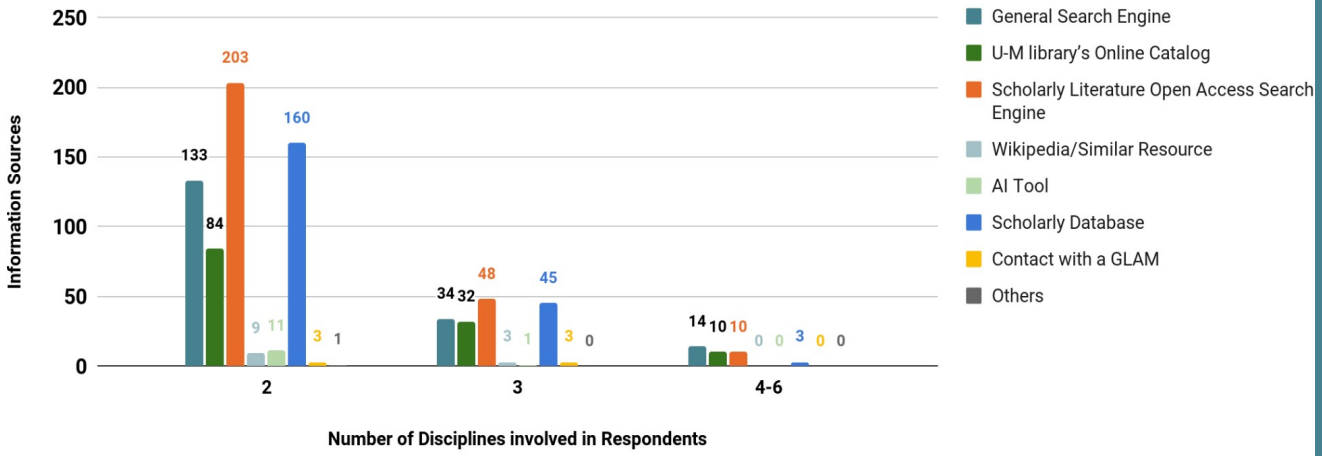


Above 75% of respondents in Reviewing Literature rate their competence in this activity's adequacy as satisfactory or better.

RQ 1. What are the common information-seeking behaviors among researchers in interdisciplinary studies in Health and STEM fields?

Research Activities: Reviewing Literature

Information Resources Background Materials



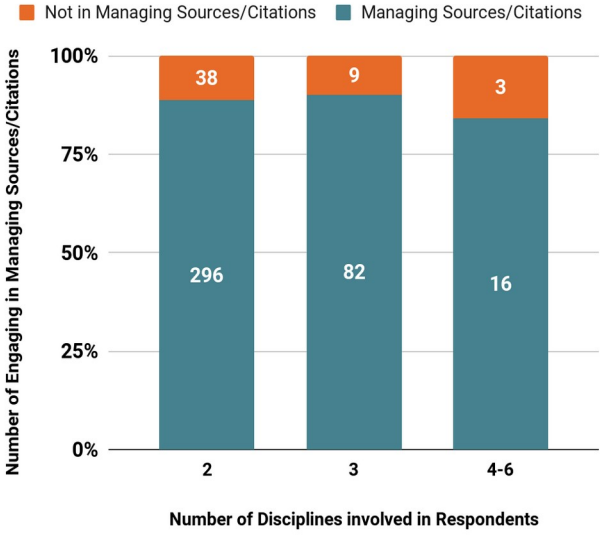
Scholarly Literature Open Access Search Engine is the most frequently used sources in majority of the interdisciplinary research.

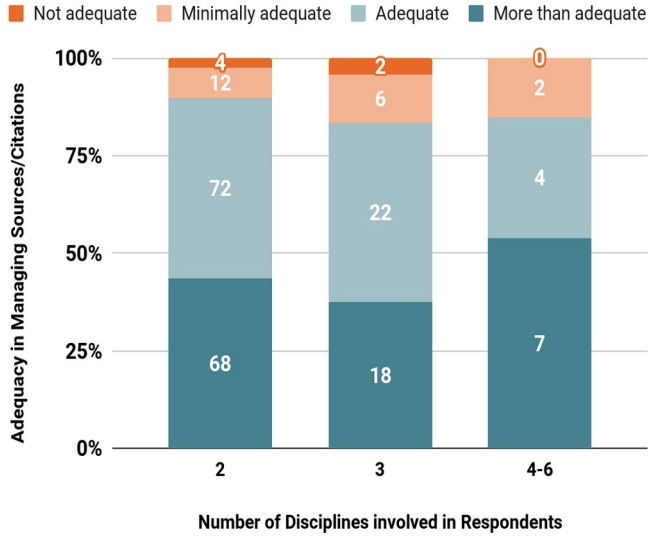
General Search Engine, U-M Library's Online Catalog, and Scholarly Database are less frequently used.

Though as the number of disciplines increases, these major resources are still being used.

RQ 1. What are the common information-seeking behaviors among researchers in interdisciplinary studies in Health and STEM fields?

Research Activities: Managing Sources/Citations

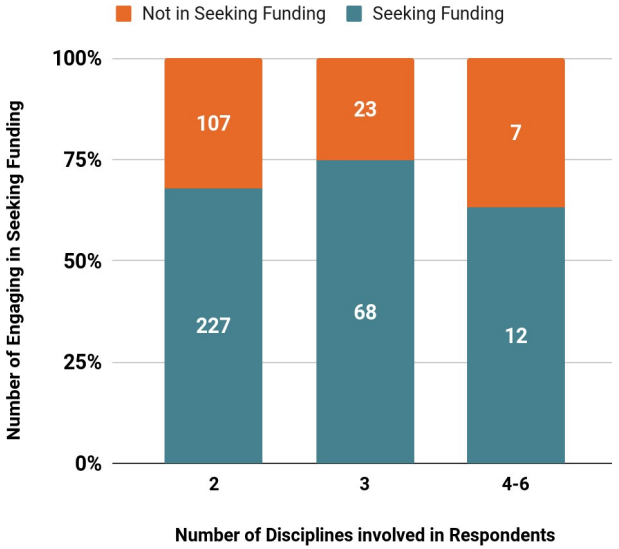


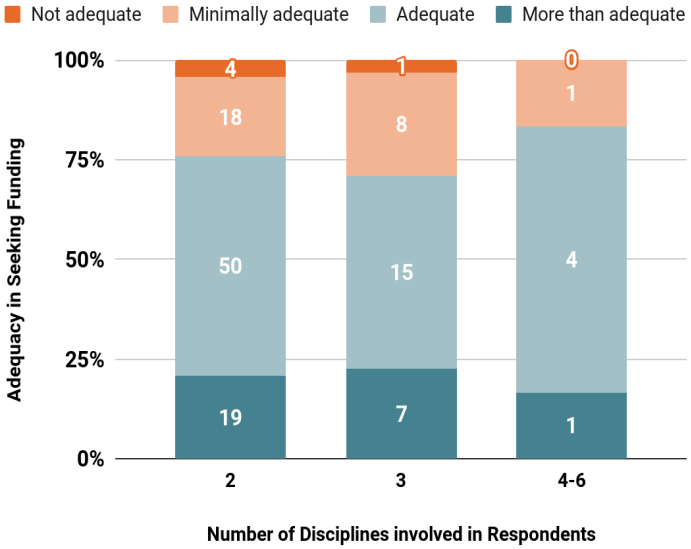


Over 75% of respondents involved in Managing Sources/Citations rate their competence in this activity's adequacy as satisfactory or better.

RQ 1. What are the common information-seeking behaviors among researchers in interdisciplinary studies in Health and STEM fields?

Research Activities: Seeking fundings

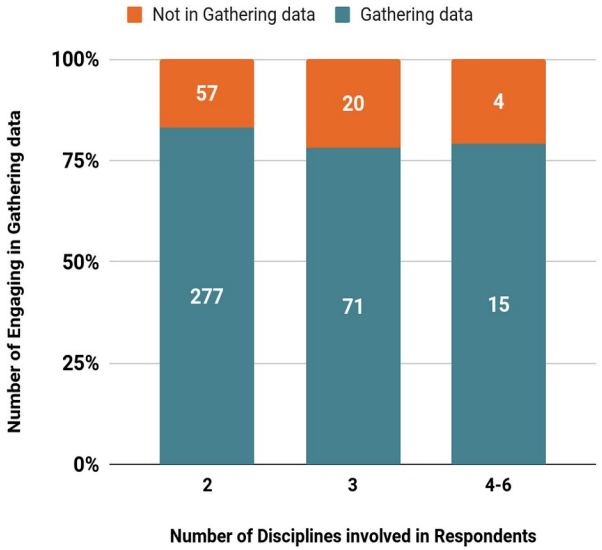


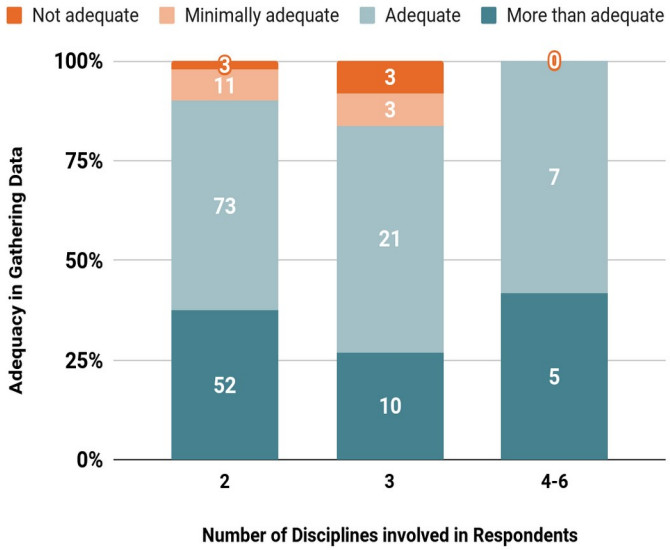


Between 50%-75% of respondents in Seeking Funding rate their competence in this activity's adequacy as satisfactory or better.

RQ 1. What are the common information-seeking behaviors among researchers in interdisciplinary studies in Health and STEM fields?

Research Activities: Gathering Data

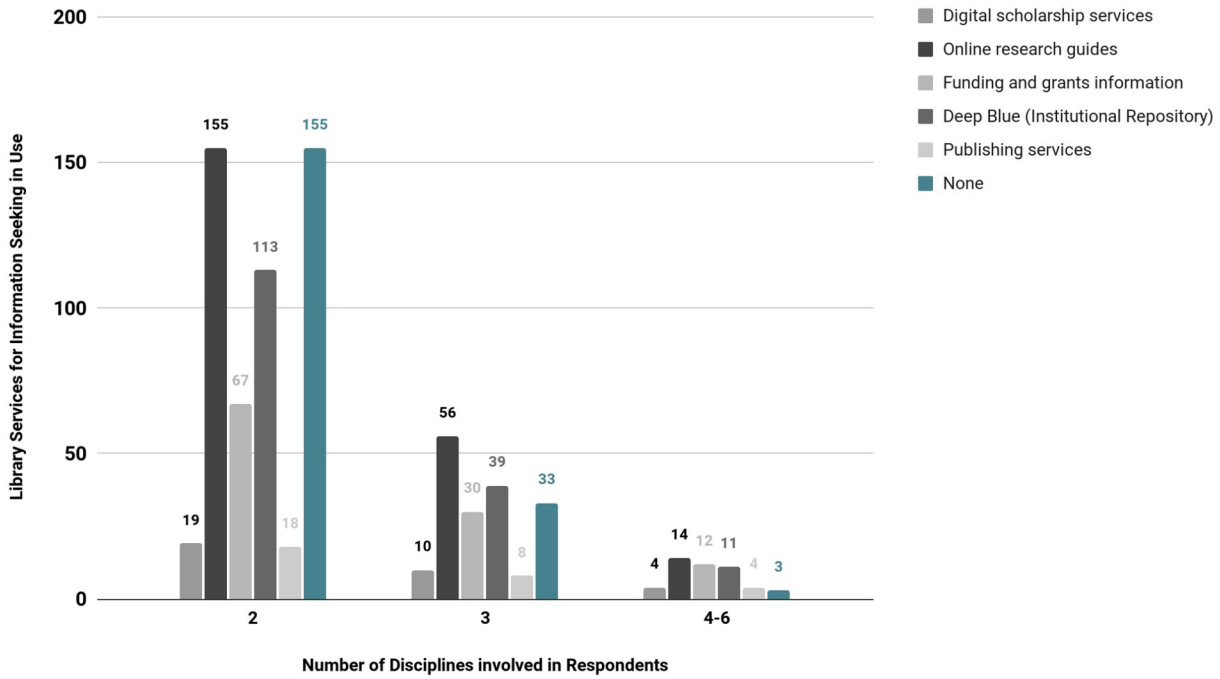




Above 75% of respondents in Gathering Data rate their competence in this activity's adequacy as satisfactory or better.

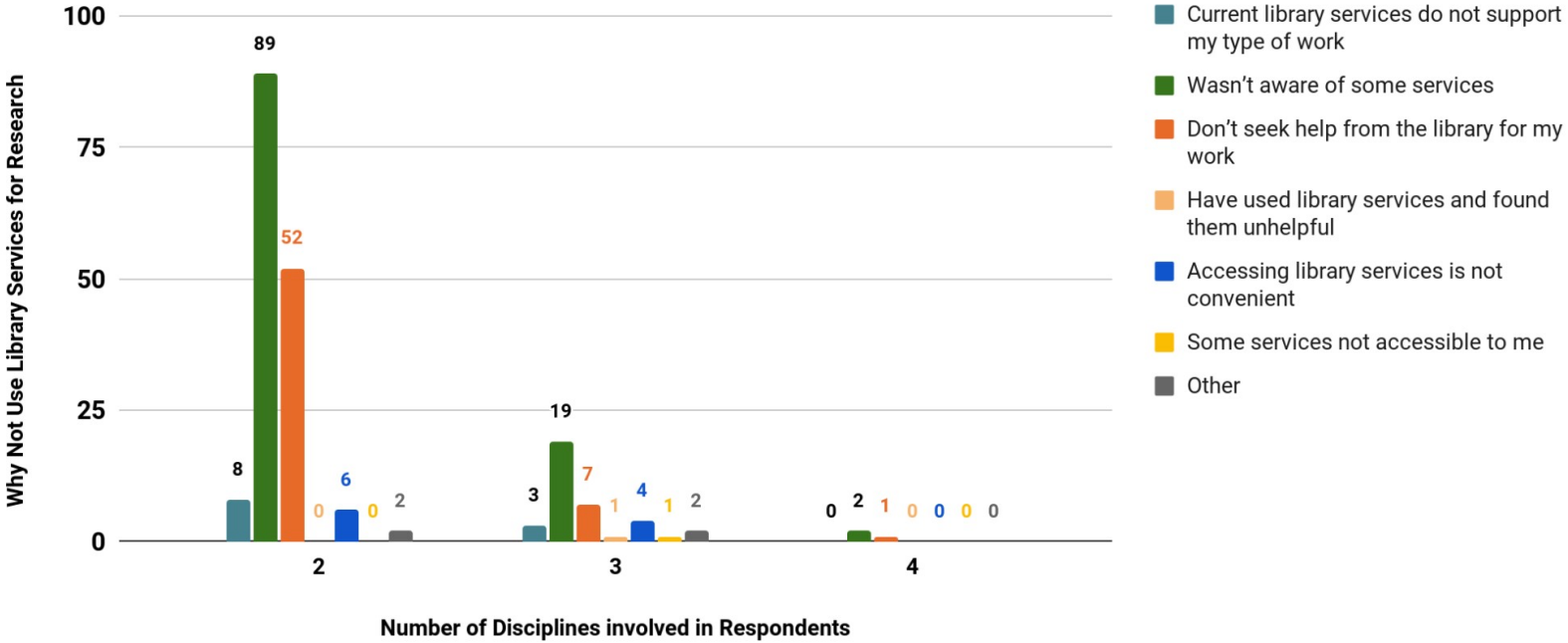
RQ 2. What issues do these researchers encounter in accessing and utilizing library resources?

Library Services Used



RQ 2. What issues do these researchers encounter in accessing and utilizing library resources?

Why not use library services?



RQ 2. What issues do these researchers encounter in accessing and utilizing library resources?

Qualitative Responses:

Awareness of services

Comment 1: "I feel like it is **hard to find information** on how I can analyze my data using the library's resources. I don't even know if the information exists. I usually go to CSCAR for analysis of my data. The campus **feels very disconnected** and I **don't know how** the library can tie into any of this."

Comment 2: "I signed up to get alerts of funding and none of the suggested funding sources apply to me. I **was unaware** that the library could assist with the other research-related tasks and **want to know more** about specifically how they can help with these."

Accessing library services is not convenient

Comment 1: "The most helpful was meeting with a librarian. The websites are **too confusing to navigate** sometimes."

Comment 2: "Since physical browsing of engineering books **has been removed**, could you please have an online browsing option where the book offerings are sorted by topic with titles listed? The **search tools are really really bad**."

RQ 3. What improvements can be made to library services to better support interdisciplinary research in these fields?

Summary

- Overall, there’s a similar trends from different numbers of interdisciplinary research respondents.
- Respondents are actively engaged in a variety of information-seeking activities, primarily reviewing literature, managing sources, gathering data, seeking funding, and sharing data.
- The library's support in research is generally seen as adequate and satisfied.
- General search engines, U-M library's online catalog, and scholarly databases are the most commonly used sources for background material.
- Improving awareness, ease to access, and demonstrating the relevance and usefulness of library services can increase the utilization.

RQ 3. What improvements can be made to library services to better support interdisciplinary research in these fields?

Recommendations

- **Awareness of services**
  Developing Clear LibGuides and tutorials
  Promoting needed services to increase awareness of services
  Hosting info session, email campaigns, and library websites

- **Accessing library services is not convenient**
  Improving search functionality
  Promoting the new web structure

- **Customized support services**
  Making workshops and personalized consultations available for respondents in higher interdisciplinary research

Challenges

- Learning SPSS and statistical methods on making analysis.
- Looking for the best approach and exam relationships across variables.

Successes

- SPSS
- Data delimitation
- Data visualization
- Data storytelling
- Communication with team and mentor

Next Steps

- Refining my current research results.
- Spending more time on literature review
- Organizing into an article, seeking opportunities for conference and publications.
- Continue to grow my network.

Question & Answer

03

THANK

This project has been funded with Federal funds from the National Library of Medicine (NLM), National Institutes of Health (NIH), under cooperative agreement number UG4LM01234 with the University of Massachusetts Chan Medical School, Lamar Soutter Library. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

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